Software Requirements Specification (SRS)

Project: Bank Management System
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Revision history

Version	Date	Author	Change summary	Approval
2.0	12-11-2026	Project Team	SRS with required structure and formatting	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Bank Management System. It defines the functional and non-functional requirements, interfaces, security objectives, security requirements, and verification criteria for the project.

The Bank Management System supports authenticated customer banking operations and authorized employee and administrator workflows. Its scope includes account and balance management, deposits, withdrawals, transaction recording and history, internal fund transfers, loan application and review, notifications, beneficiary management, profile settings, audit logging, system health monitoring, and database backup and recovery.

Developers, QA Engineers, project team members, course instructors, and assessment evaluators.

List of acronyms: SRS, FR, NFR, RBAC, CSRF, API, UI, CSV, SHA-256, RTM.

2. Overall description

The Bank Management System is a software application used by Customers, Employees, and Admins. It provides role-based access to banking operations through a user interface, server-side application logic, and a database that stores account, transaction, loan, notification, beneficiary, audit, and backup-related information.

Major product functions (detailed):

- User authentication and login
- Role-based access control for Customer, Employee, and Admin
- Account details and balance viewing
- Deposits and withdrawals
- Transaction validation and recording
- Internal fund transfers with atomic commit/rollback
- Loan application submission, review, approval, rejection, and status tracking
- Transaction history, filtering, sorting, pagination, and CSV export
- Transaction and loan notifications
- Beneficiary management
- Profile and password management

- Audit logging and authorized review/export
- System health checks and database backup/recovery

Customer: uses permitted account, balance, deposit, withdrawal, transfer, loan, transaction history, notification, beneficiary, and profile functions.
Employee: performs authorized employee operations, including permitted deposits/withdrawals and loan application review and decisions.
Admin: performs authorized administrative operations, including audit log review/export, system health checks, and backup/recovery.

Browser-based client interface, server-side application/API, and SQLite database environment as described by the project requirements. Specific supported browser versions and production deployment details are not defined in the supplied project SRS.

The system shall enforce server-side role permissions, protect authenticated sessions and state-changing requests, use integer paise for balances, process transfers atomically, and restrict audit and backup/recovery administration to authorized users.

3. External interface requirements

The system provides role-appropriate interfaces for Customer, Employee, and Admin users. Interfaces support responsive layouts, labelled controls, keyboard focus, reduced-motion behaviour, and clear feedback for user actions.

No dedicated hardware interface is specified in the Bank Management System requirements. The application is accessed through client devices using a supported browser.

The software interfaces comprise the browser-facing user interface, server-side application/API logic, and SQLite database. No external payment gateway, bank-core integration, or third-party messaging provider is specified in the supplied project requirements.

The requirements specify signed HttpOnly sessions and CSRF checks for protected interactions. The exact network protocol, TLS deployment configuration, external communication retry policy, and API endpoint catalogue are not specified in the supplied project SRS.

4. System features (detailed)

The following table lists all functional requirements for the Bank Management System. Each requirement includes its ID, description, type, priority, source/stakeholder, acceptance criteria, test case reference, and comments/dependencies.

Functional Requirements

Issue No.	Type	Story name/description	Priority	Story Points	Acceptance	Rationale
#2	Functional	Authentication & Login: Users must authenticate before accessing the system. Signed HttpOnly sessions, bcrypt password hashing, CSRF checks, and temporary lockout protect access.	High	3	Pass: Valid login creates a session; invalid credentials are rejected; CSRF is checked; 5 failed logins trigger a 15-minute lockout. Fail: Authentication/CSRF is bypassed or passwords are stored in plain text.	Authentication prevents unauthorized users from accessing the banking system.
#3	Functional	Role-Based Access Control: Server-side permissions restrict Customer, Employee, and Admin actions and prevent access to other customers' records.	High	5	Pass: Each role accesses only permitted functions and data. Fail: Restricted actions or another customer's records are accessible.	Different banking roles require different permissions to protect sensitive operations.
#4	Functional	Account & Balance Management: Customers can view account details and balances. Account numbers are unique 12-digit values; balances use integer paise and account status is tracked.	High	3	Pass: Account details and balances are accurate; account numbers are unique and status rules are enforced. Fail: Balances or account status are incorrect.	Customers need accurate account information to manage their money.
#5	Functional	Deposit Money: Customers or authorized employees can deposit a positive amount into an eligible active account.	High	3	Pass: Balance and ledger update together and confirmation is shown. Fail: Invalid deposits are accepted or balance and ledger disagree.	Deposits are a core banking operation and must be recorded accurately.
#6	Functional	Withdraw Money: Customers or authorized employees can withdraw a positive amount from an active account when sufficient funds are available.	High	3	Pass: Valid withdrawals update balance and ledger; overdrafts are rejected. Fail: An overdraft succeeds or records are inconsistent.	Withdrawals directly affect customer funds and require accurate validation.
#7	Functional	Transaction Validation & Recording: The system validates transactions and records a unique reference, type, amount, timestamp, account, and resulting balance.	High	5	Pass: Successful transactions have accurate ledger details. Fail: Invalid, duplicate, or incomplete transaction records are stored.	Validation and accurate records are necessary for reliable banking operations and auditing.
#8	Functional	Fund Transfer Between Accounts: Customers can transfer funds between eligible	High	5	Pass: Both account balances and transfer ledger entries update correctly.	Internal transfers are a core banking function and

		internal accounts after ownership, destination, account status, and balance checks.			Fail: Self-transfer, unauthorized source, invalid destination, or insufficient funds are accepted.	must move funds accurately.
#9	Functional	Atomic Transfer & Rollback: A transfer must update both accounts, ledger entries, and transfer record within one database transaction.	High	8	Pass: All changes commit together or all roll back on failure. Fail: Any partial transfer update remains after failure.	Atomic processing prevents partial transfers and protects customer balances.
#10	Functional	Loan Application: Customers can submit validated personal, home, education, vehicle, or business loan applications with a unique reference ID.	Medium	5	Pass: A valid application is saved with a reference ID. Fail: Invalid applications are accepted or submitted applications are lost.	Online loan applications make the loan process accessible and traceable.
#11	Functional	Loan Application Review: Authorized loan staff can view submitted applications, inspect details, and filter the review queue by status.	Medium	3	Pass: Authorized staff can retrieve and review applications. Fail: Unauthorized access is allowed or application details are unavailable.	Loan officers need access to application information before making decisions.
#12	Functional	Loan Approval / Rejection & Status: Authorized loan staff can approve or reject applications; rejection requires a reason and the customer can view the updated status.	Medium	5	Pass: Decision and status are saved; rejection reason is retained and customer is notified. Fail: Unauthorized or incomplete decisions are accepted.	Clear loan decisions and status tracking keep the lending process transparent.
#13	Functional	Transaction History: Customers can view chronological deposits, withdrawals, and transfers with reference IDs, types, amounts, and resulting balances.	Medium	3	Pass: The customer's history shows accurate transaction details. Fail: Records are missing, duplicated, incorrect, or expose another customer's data.	Transaction history helps customers verify activity and identify errors or unauthorized transactions.
#14	Functional	Transaction Filtering & Pagination: Customers can search, filter by transaction type and date range, sort, and navigate server-paginated history.	Low	3	Pass: Filters and pages return the correct matching records. Fail: Results contain unrelated, missing, or duplicated records.	Filtering and pagination make large transaction histories easier to search and use.
#15	Functional	Transaction Statement / Export: Customers can export account transaction statements as CSV, with spreadsheet formula-injection values sanitized.	Low	5	Pass: CSV contains correct transaction details and unsafe formulas are escaped. Fail: Export is inaccurate, incomplete, or unsafe.	Statements provide customers with a usable record for review and external purposes.
#16	Functional	Transaction & Loan Notifications: In-app notifications are generated for deposits,	Medium	5	Pass: Configured events create accurate notifications that users can read and	Timely notifications keep customers informed

		withdrawals, transfers, and loan decisions, with unread counts and mark-as-read support.			mark as read. Fail: Required notifications are missing or incorrect.	about important account and loan activity.
#17	Functional	Beneficiary Management: Customers can add, view, update, and remove eligible beneficiaries for future transfers, subject to validation and transfer limits.	Medium	5	Pass: Valid beneficiaries are saved and transfer limits are enforced. Fail: Invalid beneficiaries or over-limit transfers are accepted.	Managing beneficiaries makes repeated fund transfers easier while keeping transfer destinations organized.
#18	Functional	Profile and Account Settings: Authenticated users can update permitted name/email details and change passwords after verifying the current password.	Low	3	Pass: Profile changes are validated; password policy is enforced and password reuse is rejected. Fail: Invalid updates or unverified password changes succeed.	Allowing customers to maintain their permitted information keeps account records current.

Non-functional Requirements

Issue No.	Type	Story name/description	Priority	Story Points	Acceptance	Rationale
#19	Security	Data Security and Role Access: The system uses bcrypt password hashing, CSRF validation, server-side RBAC, security response headers, and browser cache controls.	High	8	Pass: Passwords are hashed; protected actions validate CSRF; roles and security/cache headers are enforced. Fail: Credentials or restricted data are exposed, or protections are missing.	A bank is a high-value target, so weak access control or unencrypted data is a serious risk.
#20	Performance	System Response Time: Request latency for common banking operations is measured and exposed through response timing headers for performance monitoring.	Medium	5	Pass: Core endpoints expose timing measurements and meet the defined response target under test conditions. Fail: Timing is not measured or response targets are regularly missed.	Slow response times frustrate customers and make the bank feel unreliable, even if the data is correct.
#21	Availability	System Availability and Reliability: Liveness/readiness checks, database diagnostics, SQLite busy-timeout	Medium	8	Pass: Health checks report service/database status and failed requests leave no open	Banking functions need to be dependable because customers

		handling, and safe request rollback support service reliability.			transactions or locks. Fail: Database errors crash the API or leave uncommitted work.	may need access to their accounts and transactions at any time.
#22	Auditability	Audit Logging: Important authentication, account, transaction, loan, and administrative events are recorded for authorized administrative review and export.	High	5	Pass: Supported events record actor, action, time, and details; audit records resist unauthorized changes. Fail: Critical actions cannot be traced or logs can be altered without authorization.	Audit records support accountability, investigation, and verification of banking activities.
#23	Usability	User Interface and Accessibility: Customer, Employee, and Admin interfaces provide responsive layouts, labelled controls, keyboard focus, reduced-motion support, and clear feedback.	Low	5	Pass: Common tasks work across screen sizes and keyboard navigation; controls and feedback are accessible. Fail: Controls lack labels/focus or layouts become unusable on narrow screens.	A clear interface reduces user errors and makes banking functions easier to use.
#24	Recovery	Backup and Recovery: Authorized admins can create online SQLite backups, verify SHA-256 checksums and database integrity, and restore after a safety snapshot.	Medium	8	Pass: Valid backups pass integrity checks and can be restored after a pre-restore snapshot. Fail: Corrupt backups are restored or unauthorized restores are allowed.	Reliable backups reduce the risk of permanent data loss and help the bank recover from failures.

5.1. Security

5.1.1 Security Objectives

SO-01 – Confidentiality: Protect customer credentials, account information, and personal/financial records from unauthorized access or disclosure.
SO-02 – Integrity and accountability: Preserve accurate and consistent banking records and ensure important actions can be traced to authorized actors.

5.1.2 Security Requirements

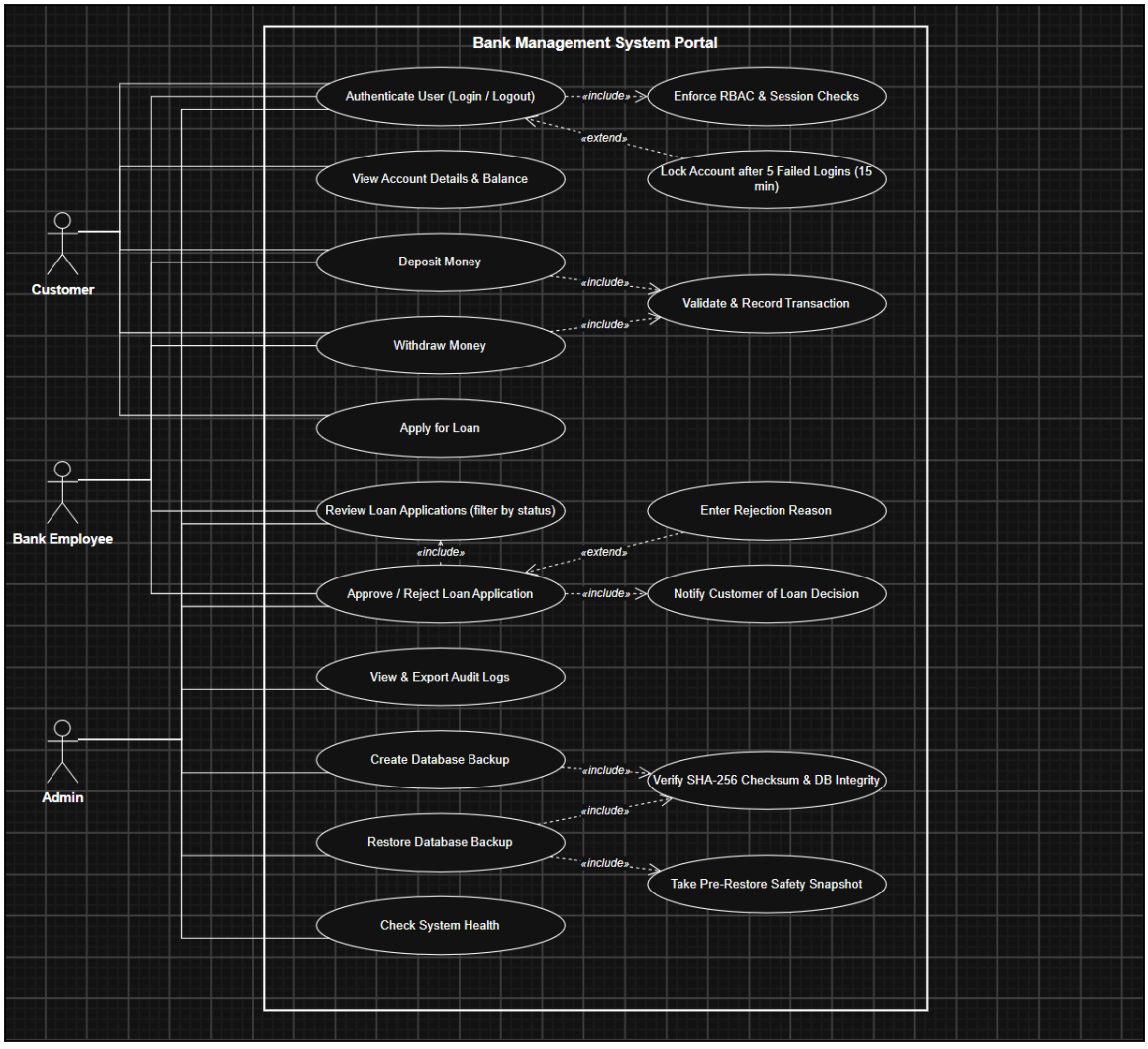
Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SR-001	The system shall use signed HttpOnly sessions and bcrypt password hashing for user authentication.	Authentication Security	High	Valid login creates a session; invalid credentials are rejected; stored passwords are hashed. Test: TC-SR-001. Source: Issues #2 and #19.
SR-002	The system shall validate CSRF protection for protected state-changing requests.	Request Security	High	Protected requests with missing or invalid CSRF validation are rejected. Test: TC-SR-002. Source: Issues #2 and #19.
SR-003	The system shall enforce server-side role-based access control for Customer, Employee, and Admin actions and customer records.	Authorization	High	Each role accesses only permitted functions and data; cross-customer record access is denied. Test: TC-SR-003. Source: Issues #3 and #19.
SR-004	The system shall apply the security response headers and browser cache controls specified by the project security requirement.	Data Protection	Medium	Applicable responses include the required security headers and cache controls. Test: TC-SR-004. Source: Issue #19.
SR-005	The system shall record important authentication, account, transaction, loan, and administrative events for authorized administrative review and export.	Auditability	Medium	Supported events record actor, action, time, and details; unauthorized users cannot alter or access audit records. Test: TC-SR-005. Source: Issue #22.

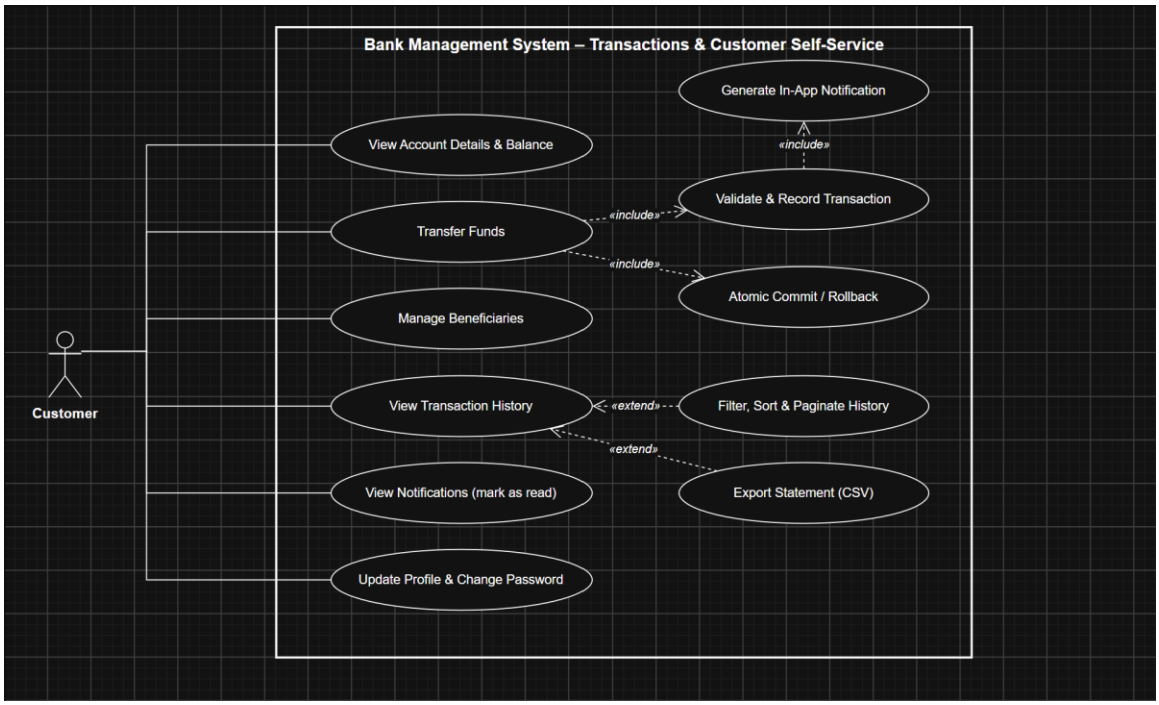
6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements are implemented and verified, no critical NFR/security failures remain, and the RTM contains the execution status and evidence for the linked test cases.
- Acceptance test suites: Authentication, role-based access control, account and balance management, deposits, withdrawals, transaction validation, transfers and rollback, loans, transaction history and export, notifications, beneficiaries, profile settings, performance, availability/reliability, audit logging, usability/accessibility, and backup/recovery.

7. System models and diagrams

7.1 UML Use-Case diagrams





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
FR-002	Authentication & Login	4.1	Authentication	TC-FR-002	N	Source issue #2; update after test execution.
FR-003	Role-Based Access Control	4.1	Access Control	TC-FR-003	N	Source issue #3; update after test execution.
FR-004	Account & Balance Management	4.2	Account Management	TC-FR-004	N	Source issue #4; update after test execution.
FR-005	Deposit Money	4.2	Deposits	TC-FR-005	N	Source issue #5; update after test execution.
FR-006	Withdraw Money	4.2	Withdrawals	TC-FR-006	N	Source issue #6; update after test execution.
FR-007	Transaction Validation & Recording	4.2	Transaction Recording	TC-FR-007	N	Source issue #7; update after test execution.
FR-008	Fund Transfer Between Accounts	4.3	Fund Transfers	TC-FR-008	N	Source issue #8; update after test execution.

FR-009	Atomic Transfer & Rollback	4.3	Atomic Transfer	TC-FR-009	N	Source issue #9; update after test execution.
FR-010	Loan Application	4.4	Loan Application	TC-FR-010	N	Source issue #10; update after test execution.
FR-011	Loan Application Review	4.4	Loan Review	TC-FR-011	N	Source issue #11; update after test execution.
FR-012	Loan Approval / Rejection & Status	4.4	Loan Decisions	TC-FR-012	N	Source issue #12; update after test execution.
FR-013	Transaction History	4.2	Transaction History	TC-FR-013	N	Source issue #13; update after test execution.
FR-014	Transaction Filtering & Pagination	4.2	History Filtering	TC-FR-014	N	Source issue #14; update after test execution.
FR-015	Transaction Statement / Export	4.2	Statement Export	TC-FR-015	N	Source issue #15; update after test execution.
FR-016	Transaction & Loan Notifications	4.5	Notifications	TC-FR-016	N	Source issue #16; update after test execution.
FR-017	Beneficiary Management	4.3	Beneficiaries	TC-FR-017	N	Source issue #17; update after test execution.
FR-018	Profile and Account Settings	4.5	Profile Settings	TC-FR-018	N	Source issue #18; update after test execution.
NFR-019	Data Security and Role Access	5	Data Security	TC-NFR-019	N	Source issue #19; update after test execution.
NFR-020	System Response Time	5	Performance	TC-NFR-020	N	Source issue #20; update after test execution.
NFR-021	System Availability and Reliability	5	Availability / Reliability	TC-NFR-021	N	Source issue #21; update after test execution.
NFR-022	Audit Logging	5	Audit Logging	TC-NFR-022	N	Source issue #22; update after test execution.
NFR-023	User Interface and Accessibility	5	UI / Accessibility	TC-NFR-023	N	Source issue #23; update after test execution.
NFR-024	Backup and Recovery	5	Backup / Recovery	TC-NFR-024	N	Source issue #24; update after test execution.
SR-001	Security requirement 1	5.1.2	Security	TC-SR-001	N	Derived from project security controls.

SR-002	Security requirement 2	5.1.2	Security	TC-SR-002	N	Derived from project security controls.
SR-003	Security requirement 3	5.1.2	Security	TC-SR-003	N	Derived from project security controls.
SR-004	Security requirement 4	5.1.2	Security	TC-SR-004	N	Derived from project security controls.
SR-005	Security requirement 5	5.1.2	Security	TC-SR-005	N	Derived from project security controls.